

UAV Flight Control System

System Test Report

Version: 1.0 | Classification: UNCLASSIFIED

Date: 2026-05-15

Project: VerificationBasedMBSE

1 Test Objectives

Verify FeedbackControl model behaviour against all applicable requirements using Simulink Test Manager suites.

Suite	Objective
FCS Communications	Verify MAVLink/Custom comms variant behaviour
Autonomous Flight	Verify waypoint navigation end-to-end
Flight Control	Verify PID control bounds during hover
Guidance Verification	Verify altitude saturation handling
AirVehicle Verification	Verify AirVehicle performance bounds
Fault Injection	Verify FaultDetector anomaly flags on injected faults
SDV_MCDC_AutoGenerated	Verify SDV-generated MCDC test cases pass
Coverage_Gaps	Verify NearWP, ModeAnomaly, NaN, Thrust, re-entry paths

2 Test Environment

Parameter	Value
MATLAB	R2026a
Simulink	R2026a
Simulink Test	v26.1 (R2026a)
OS	Windows 11 Enterprise
Solver	Fixed-step discrete Ts=0.01s
Stop Times	5s-15s per TC

3 Test Cases Executed

Suite	Test Case	Model
FCS Communications	MAVLink Spoof	QuadArchitecture
FCS Communications	MAVLink Normal Operation	QuadArchitecture
Autonomous Flight	Waypoint Navigation	QuadArchitecture
Flight Control	FCS Hover Control Bounds	QuadArchitecture
Guidance Verification	Guidance Altitude Saturation	Guidance
AirVehicle Verification	AirVehicle Performance Bounds	QuadArchitecture
Fault Injection	TC_FaultA_IMU_Loss_Recovery	FeedbackControl
Fault Injection	TC_FaultB_AltBias_Detection	FeedbackControl
SDV_MCDC_AutoGenerated	SDV_TC1_32obj	FeedbackControl
SDV_MCDC_AutoGenerated	SDV_TC2_3obj	FeedbackControl
SDV_MCDC_AutoGenerated	SDV_TC3_4obj	FeedbackControl

SDV_MCDC_AutoGenerated	SDV_TC4_2obj	FeedbackControl
Coverage_Gaps	TC_COV1_WaypointProximity	FeedbackControl
Coverage_Gaps	TC_COV2_ModeAnomaly	FeedbackControl
Coverage_Gaps	TC_COV3_RelPosNaN	FeedbackControl
Coverage_Gaps	TC_COV4_ThrustOvershoot	FeedbackControl
Coverage_Gaps	TC_COV5_NearWP_Reentry	FeedbackControl

4 Pass / Fail Results

All 17 test cases passed. 0 failures. Results recorded 2026-05-15.

Suite	Total	Pass	Fail
FCS Communications	2	2	0
Autonomous Flight	1	1	0
Flight Control	1	1	0
Guidance Verification	1	1	0
AirVehicle Verification	1	1	0
Fault Injection	2	2	0
SDV_MCDC_AutoGenerated	4	4	0
Coverage_Gaps	5	5	0
TOTAL	17	17	0

5 Coverage Achieved

Metric	Achieved	Target	Status
Decision	93% (14/15)	85%	MET
Condition	100% (14/14)	100%	MET
MCDC	100% (7/7)	100%	MET
Execution	100% (30/30)	100%	MET
Chart Decision	100% (9/9)	100%	MET

6 Fault Injection Results

Test Case	Fault Trigger	Detection	Result
TC_FaultA_IMU_Loss_Recovery	WorldPos z>0 sustained	ThrustAnomaly (Port 1)	Pass
TC_FaultB_AltBias_Detection	WorldPos z-step at t=3s	PositionAnomaly (Port 3)	Pass
TC_COV2_ModeAnomaly	Mode=uint8(5) injection	ModeAnomaly (Port 2)	Pass
TC_COV3_RelPosNaN	WorldPos=NaN	PositionAnomaly (Port 3)	Pass

7 Anomalies

None identified during test execution. All assertions met.

8 Conclusion

All 17 test objectives met. The UAV FCS FeedbackControl model is verified against requirements with >85% decision coverage (achieved 93%) and 100% MCDC coverage. GuidanceCompute/Chart achieved 100% decision coverage. Fault detection verified for all 4 injected fault scenarios. System is ready for DO-178C Stage of Involvement review.